

Assignment → Internal → 1

1. Define Agile manifesto and list its four core values.

→ The Agile manifesto is a document created in 2001 by 17 software developers. It provides a better way to develop software by focusing on teamwork, customer satisfaction, flexibility and continuous ~~improvement~~ improvement.

(I) Individuals and Interactions over Processes and tools.

(II) Working software over Comprehensive Documentation

(III) Customer Collaboration over Contract Negotiation.

(IV) Responding to change over following a plan.

2. What are the three ways of DevOps? Explain each briefly.

⇒ The flow → It focuses on development, testing, deployment then Customer.

→ The main objective is to reduce the error, automate the work flow & faster delivery.

⇒ Feedback → It's ~~focuses~~ creates rapid feedback tools which contains operation then development & then improvement.

→ It's benefits by fixing the bug faster and improve reliability.

⇒ Continue learning → It encourages experimentation, innovation and Continuous learning.

3. List the roles involved in scrum framework.

→ Scrum is one of the most popular Agile framework

used to develop software in short iterations called sprints.

(I) Project owner → Defines Product requirements
→ Maintains and Prioritizes the Product Backlog.
→ Represents Customer needs.

(II) Scrum master → Guides the scrum team.
→ Removes obstacles faced by the team.
→ Ensure Scrum Practices are followed.

(III) Development Team → Designs, develops, tests and delivers the Product.
→ Works together to complete Sprint goals.
→ Responsible for Product Quality.

4. What is version Control? state its advantages.

→ version Control is a system that records changes made to files or source code over time.

→ It allows developers to track, manage and restore previous versions whenever required.

⇒ Advantages: →

1) keeps complete history of code changes.

2) Allows multiple developers to work together.

3) makes it easy to restore previous versions.

4) Prevents loss of source code.

5) Helps identify and fix errors quickly.

6) Supports Collaboration and teamwork.

Ex → Git, GitLab, GitHub

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5) What is Continuous Integration (CI)? Name any two CI tools.

→ Continuous integration is a DevOps Practice where developers frequently merge their code into a shared repository. Every code change is automatically built and tested to detect errors early.

⇒ Benefits :-> Finds bugs early

:-> Improves Software Quality

:-> Reduces integration Problems.

⇒ Two CI tools :-> GitLab CI/CD

:-> Github Actions

:-> Travis CI

6) Explain the relationship between Agile methodology and DevOps.

→ Agile is software development methodology that focuses on developing software quickly with customer feedback.

→ DevOps is a culture that combines Development and Operation to automate software building, testing and deployment.

⇒ Relationship

→ Agile focuses on developing software quickly, while DevOps focuses on delivering software quickly and reliably.

→ Agile improves collaboration between developers and customers whereas DevOps improves collaboration between developers and operations teams.

→ Agile delivers software in small iterations.

→ Both aim to improve software quality, speed and customer satisfaction.

7. Differentiate between Scrum and Kanban with suitable examples.

Scrum

Kanban

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| → Work is divided into fixed Sprints. | → Work flows continuously without Sprints. |
| → Roles like Product owner and Scrum master are defined. | → Not fixed roles are required. |
| → Tasks are planned before each Sprint. | → New tasks can be added anytime. |
| → Progress is measured after every Sprint. | → Progress is tracked continuously using a Kanban Board. |
| → Best for Projects with changing requirements. | → Best for Continuous Support and maintenance work. |

8. Explain the Purpose of user stories and story Points in Agile development.

→ A user story is a short description of a feature written from the user's point of view. It explains what the user wants and why.

⇒ Purpose → Understand Customer requirements.

→ Focus on user needs.

→ Improve Communication.

→ Help in Planning Sprints.

⇒ Story Point → Story Points are used to estimate the effort, complexity and time required to complete a user story.

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- Purpose → Estimate work accurately.
 - Plan sprint workload.
 - Improve team productivity.

9) Describe the workflow of Git branching and merging.

- Git branching allows developers to work on new features without affecting the main Project - merging combines completed changes back into the main branch.

⇒ Git branching and merging workflow: ⇒

- 1) Create a new branch from the main branch.
- 2) Develop the new feature in that branch
- 3) Save changes using commits.
- 4) Test the feature.
- 5) Merge the feature branch into main branch.
- 6) Delete the feature branch if no longer needed.

- ⇒ Benefits: ⇒ multiple developers can work simultaneously
- ⇒ keeps the main branch stable.
 - ⇒ Easy Collaboration.
 - ⇒ Reduces conflicts.

Ex: → A developer creates a branch named login-feature, completes the login page, tests it and merges it into the main branch.

10) Explain how Continuous Integration improves software quality.

- Continuous Integration is a DevOps practice where developers frequently merge their code into a shared repository - Every change is automatically built and tested.

⇒ CI Improves Software Quality:

- 1> Detects bugs early through automatic testing
- 2> Reduces integration Problems.
- 3> Ensures every code changes is tested.
- 4> Improves Code quality with frequent builds.
- 5> Enables faster bug fixing
- 6> supports faster and more reliable software delivery.

⇒ Benefits of CI: → Better Software Quality

→ Faster development

→ Reduced manual testing

→ Increased team Productivity.

⇒ Ex CI Tools: → Github Actions

→ Gitlab CI/CD

11> A software team needs to manage frequent code changes from multiple developers. Demonstrate how Git branching and Pull Requests can be used to avoid conflicts.

→ When multiple developers work on the same project, Git branching and Pull Requests help manage code changes without creating conflicts. Each developer creates a separate branch to work on a new feature or bug fix. This keeps the main branch safe and stable.

S1> Create a separate branch for each feature.

S2> Each developer works only on their own branch.

S3> Commit and Push changes to GITHUB.

S4> Create a Pull Request.

S5> Team members review the code.

S6> After approval, merge the branch into main branch.

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- ⇒ Advantages :- Prevents Code Conflicts
 - :- Improves Collaboration
 - :- Easy code review before merging.

12) A startup wants to adopt Agile for Product development. Illustrate how Scrum Ceremonies can be organized during a two-week Sprint.

→ A two-week Sprint is a fixed Period in Scrum during which the team develops a set of features. Scrum Ceremonies help the team stay organized and complete the work successfully.

⇒ Scrum Ceremonies :-

- 1) Sprint Planning :- The team selects tasks from the Product Backlog and Plans the sprint.
- 2) Daily Scrum :- A 15-minute daily meeting to discuss, Plans and Issues.
- 3) Sprint Review :- At the end of the Sprint, the team demonstrates the completed work to Stakeholders.
- 4) Sprint Retrospective :- The team discusses what well, what Problems occurred and how to improve in the next Sprint.

- ⇒ Benefits :- Better Communication
- :- Regular Progress tracking
 - :- Foster Problem Solving
 - :- Continuous improvement.

13) Given a Project backlog, estimate Story Points for five user stories using Fibonacci estimation.

→ story Points are used in Agile to estimate the effort needed to complete a user story.

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→ The fibonacci sequence (1, 2, 3, 5, 8, 13) is commonly used because it reflects increasing complexity.

User story	story Point
User login →	2
User Registration →	3
Forgot Password →	5
Online Payment →	8
Admin Dashboard →	13

⇒ Benefits of story points → Helps estimate work accurately
→ Improves Sprint Planning
→ Measures task complexity.

14) Apply Kanban Principles to Organize the workflow of an online shopping application development team.

→ Kanban is an Agile method that helps teams manage work visually using a kanban board. The work is divided into different stages so everyone can track progress easily.

⇒ workflow →

To do → New tasks such as login, Product Page, cart & Payment.
To Progress → Tasks currently being developed.

Testing → Completed tasks are tested for bugs.

Done → Successfully tested tasks are completed.

⇒ Benefits → Easy task tracking

Better teamwork

Faster delivery

Reduced work delays.

15) Configure a basic CI workflow using Github Actions or Jenkins for a Java Project that automatically builds and executes unit tests.

→ A Continuous Integration (CI) workflow automatically builds and tests a Java Project whenever code is pushed to the repository. This helps detect errors early and improves software quality.

⇒ workflow:→

- 1) Developer pushes code to Github.
- 2) Github Actions or Jenkins detects the change.
- 3) The Java Project is automatically built.
- 4) Unit tests are executed.
- 5) Test results are displayed.
- 6) If all tests pass, the code is ready for deployment.

⇒ Benefits:→ Automatic code building
→ Early bug detection
→ Better software quality.
→ faster Project delivery.

16) Compare Agile and Traditional waterfall methodologies based on flexibility, customer involvement and Project delivery.

Agile

waterfall

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| → Flexible and accepts changes | → Fixed Plan with limited changes |
| → Customer is involved throughout the Project | → Customer involvement is mostly at the beginning and end. |

- Continuous testing during development.
- Testing is done after development is completed.
- Best suitable for Projects with changing requirements.
- Best suitable for Projects with fixed and clear requirements.
- ⇒ Advantages:→
 - faster delivery.
 - Better customer satisfaction.
- ⇒ Advantages:→
 - Simple to understand.
 - Well-Planned documentation.

17) Analyze the reasons behind the failure of DevOps implementation in an organization.

- DevOps implementation may fail if an organization does not follow proper planning and teamwork, successful DevOps requires collaboration, automation and continuous improvement.

⇒ Reasons for failure:→

- Poor Communication between development and operation teams.
- Lack of management support.
- Resistance to change.
- Insufficient training and skills.
- Limited automation.
- Poor Planning and unclear goals.

⇒ Solutions:→ Improve team collaboration

→ Provide proper training

→ Automate testing and ~~dev~~ deployment.

→ set clear objectives.

18) Compare Git flow and Github flow branching strategies for enterprise software development.

Git Flow

GitHub flow

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| → uses multiple branches like main, develop, feature, release. | → uses only main feature branches. |
| → suitable for large and complex projects. | → suitable for small and fast-moving projects. |
| → supports scheduled releases. | → supports continuous deployment. |
| → more complex workflow. | → simple and easy workflow. |
| → Better for enterprise software development. | → Better for web applications and cloud projects. |

19) Analyze the impact of code reviews and Protected Branches on software quality.

⇒ Code review is the process of checking code before merging it into the main branch.

⇒ Protected Branches prevent direct changes to important branches like main.

⇒ Impact on software quality:→

- 1) Finds bugs before deployment
- 2) Improves code quality and readability.
- 3) Encourages coding standards.
- 4) Prevents accidental deletion or modification of important code.
- 5) Ensures only reviewed and approved code is merged.
- 6) Improves collaboration among developers.

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20) Examine the relationship between velocity, SPrint Planning and Release Planning in Scrum.

→ In scrum, velocity, SPrint Planning and Releases Planning work together to manage Project Progress.

⇒ Velocity is the amount of work completed by a scrum team during one SPrint. It is measured using Story Points.

⇒ SPrint Planning uses the team's velocity to decide how much work can be completed in the next SPrint.

⇒ Release Planning uses the average velocity of several sprints to estimate when the complete product will be ready for release.

⇒ Relationship: → velocity measures team performance.

→ SPrint Planning uses velocity to plan SPrint tasks.

→ Release Planning uses velocity to estimate delivery dates.

21) Evaluate whether Scrum or Kanban is more suitable for maintaining an e-commerce application.

⇒ Scrum is an Agile framework that works in fixed SPrint.

⇒ Kanban is an Agile method that manages work through a continuous workflow.

→ for maintaining an e-commerce application, Kanban is more suitable becoz maintenance work is continuous and unpredictable.

⇒ Justification: →

1. New issues and customer requests can be handled immediately.
2. No need to wait for the next sprint.
3. Tasks are managed using a Kanban Board.
4. Continuous workflow improves response time.
5. Better for bug fixing, updates and support.

Ex ⇒ If customers report payment or login issues, the development team can fix them immediately using Kanban without waiting for a sprint.

22) An organization performs manual testing after development. Evaluate the benefits and challenges of introducing Continuous Integration.

→ Continuous Integration automatically builds and tests software whenever developers make code changes. Replacing only manual testing with CI improves software quality and development speed.

⇒ Benefits: → Detects bugs early.

→ Reduces manual testing effort.

→ Improves code quality.

→ Saves time and cost.

→ Speeds up software delivery.

⇒ Challenges: → Initial setup requires time and resources.

→ Team members need proper training.

→ Automation tools may require maintenance.

→ Existing workflows may need changes.

23) Evaluate different branching strategies (Git flow, feature Branching, Trunk-Based Development) for a large-scale software Project.

→ Different branching strategies are used depending on the Project size and development Process.

⇒ Git Flow → Best for large Projects.

→ uses feature, develop, release and main branches.

→ Suitable for planned releases.

⇒ Feature Branching → Each new feature is developed in its own branch.

→ Easy to review and test.

→ Reduces Conflicts.

⇒ Trunk-Based Development → Developers frequently merge changes into the main branch.

→ Supports Continuous integration

→ Best for fast development teams.

24) Assess the effectiveness of retrospectives in improving Agile team performance.

→ A Sprint Retrospective is a scrum meeting held after every sprint. During this meeting, the team discusses what went well, what problem occurred and how to improve in the next sprint.

⇒ Benefits → Improves teamwork

→ Identifies Project Problems.

→ Encourages open communication

→ Increases Productivity.

→ Supports Continuous improvement.

25) Evaluate the importance of Artifact Repositories such as Nexus in modern CI/CD pipelines.

→ An Artifact Repository is a storage system used to manage software packages, libraries and build files. Nexus Repository is a popular artifact repository used in DevOps and CI/CD pipelines.

⇒ Importance of Nexus:

- stores build artifacts safely.
- makes software dependencies easy to manage.
- Supports faster deployments.
- Improves security and version management.

⇒ Benefits:

- Centralized storage
- Easy artifact sharing
- Better software reliability.
- Faster release process.

26) Design an Agile development lifecycle for a library management system using Scrum.

→ A library management system can be developed using the Scrum framework, where the project is divided into small sprints. Each sprint delivers a working feature of the system.

→ The Product owner defines the requirements, the Scrum master manages the process and the Development team builds the software.

⇒ Lifecycle:

- create the Product Backlog.

- Plan the Sprint

- Develop features such as login, Book search, Book Issues and Return.

- Test the Completed features.
- Conduct Sprint Review and ~~Retrospective~~ Retrospective.
- Repeat the Process until the Project is Completed.

27) Create a Git workflow for a team of team of ten developers that includes feature branches, Pull requests, Protected branches and release branches.

→ Git workflow →

1. Create a main branch for stable code.
2. Each developer creates a separate feature branch.
3. Develop and test new features.
4. Push code and create a Pull request.
5. Team members review the code.
6. merge approved code into the develop branch.
7. create a release branch before deployment.
8. Protect the main branch so direct changes are not allowed.
9. merge the release branch into the main branch after final testing.

28) Design a Jenkins or Github Actions CI Pipeline that builds tests and Packages a Spring Boot application.

→ A CI Pipeline automates the Process of building, testing and Packaging a Spring Boot application whenever code is Pushed to the repository.

⇒ CI Pipeline Steps →

1. Developer Pushes code to Github
2. Github Actions or Jenkins starts automatically.
3. Download Project dependencies.
4. Build the spring Boot application
5. Run unit tests.

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6. Package the application into a JAR file.
 7. Store the build artifact for deployment.

29) Develop a release planning strategy for a mobile banking application using Agile Principles and velocity Estimation.

- A release planning strategy helps organize and deliver features for a mobile banking application.
- The team's velocity is used to estimate how much work can be completed in each sprint.

⇒ Release Planning steps :-

- 1) Create and Prioritize the Product Backlog.
- 2) Estimate story points for each feature.
- 3) Plan sprint tasks based on team velocity.
- 4) Review completed work after every sprint.
- 5) Release tested features to users.
- 6) Collect customer feedback and improve future releases.

⇒ Benefits :- Better Planning

→ Faster delivery

→ Higher Customer satisfaction.

30) Propose a complete DevOps adoption roadmap for a software company transitioning from a traditional development process.

- A devops adoption roadmap helps an organization move from traditional software development to a modern DevOps approach.

⇒ Roadmap :-

- 1) Assess the current development process.

- 2> Train employees in Agile and DevOps Practices.
- 3> Adopt version control using Git.
- 4> Implement Continuous Integration.
- 5> Automate testing and deployment.
- 6> Use monitoring and feedback tools.
- 7> Continuously improve the development Process.

⇒ Benefits → Faster software delivery
⇒ Better collaboration
⇒ Improved software quality.
⇒ Reduced manual work.
⇒ Higher customer satisfaction.

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